Introduction to meteorology: atmosphere and trees

Supplement BG08 · introductory draft

What is weather?

Weather is the state of the atmosphere at a place and time. Temperature, pressure, wind, humidity, precipitation and solar radiation describe different aspects. Meteorology studies these conditions and their changes, using observations and forecasts.

Observed variables

Air temperature is commonly reported in °C, pressure in hPa, wind speed in m/s and precipitation as a depth in mm over a specified period. Wind direction names the direction from which wind comes. Relative humidity is the ratio of actual to saturation water-vapour pressure at the same temperature, normally in percent.

Trees experience radiation, wind, water and temperature together. Sensor height, terrain and surroundings matter, and a station need not represent conditions inside a forest.

Pressure and wind

Air has mass, and gravity contributes to atmospheric pressure. Station pressure and pressure reduced to sea level are different quantities. Compare like with like on a map.

Why wind changes

A pressure gradient exerts a force from higher toward lower pressure. At large scales the Coriolis effect associated with Earth's rotation also matters. Near the surface friction and terrain modify both speed and direction. Winds do not always cross isobars at right angles.

Why this matters for trees

Wind affects leaf water exchange and temperature. Strong winds may break branches or uproot trees. Ring width alone cannot establish that wind damage occurred.

Heating or cooling near the surface changes air density and can contribute to rising or sinking motion. Rising air expands as pressure decreases and cools; water vapour may condense to form clouds. Rising motion does not guarantee clouds or precipitation.

Radiation and the water cycle

Incoming solar shortwave radiation warms leaves and ground; land and atmosphere also emit longwave radiation. Absorption, reflection and emission, together with heat and moisture exchange, affect temperature. Clouds and canopy shade alter incoming radiation.

Moving water

Water moves through evaporation, condensation, precipitation, runoff and infiltration. Evapotranspiration combines surface evaporation and plant transpiration. Equal precipitation totals need not imply equal water availability: timing, soil and evapotranspiration matter.

Precipitation is usually expressed as water depth over a specified period. Distinguish snow depth from precipitation expressed as liquid-water equivalent.

Transpiration depends on the vapour gradient between leaf and air, stomatal opening and wind. When water uptake cannot keep pace, stomata may close to reduce water loss. Growth responses vary with species, soil and season.

Seasons and atmospheric circulation

Solar angle and day length vary with season and latitude, changing how the surface is heated. Uneven heating, Earth's rotation and the distribution of land and oceans contribute to atmospheric circulation. In mid-latitudes, upper-air westerlies influence the movement of weather systems.

Land and sea

Land and ocean warm and cool differently. This contributes to seasonal differences in pressure patterns and winds. Monsoons involve seasonal changes in winds and precipitation over broad regions, with distinct local and topographic effects.

Conditions at a forest also depend on elevation, slope and canopy structure. When using a regional climate index, check how it relates to the forest site.

Soil water and tree response

Not all precipitation reaches roots. Some is intercepted and evaporates from the canopy, some runs off, and some infiltrates into soil. Water available to roots depends on soil properties and depth as well as root distribution.

Atmospheric dryness and transpiration

Warmer air has a higher saturation vapour pressure. Even at the same relative humidity, the difference between saturation and actual vapour pressure changes with temperature. This vapour pressure deficit helps describe the atmosphere's capacity to receive water vapour.

Transpiration and ring width are not determined by this deficit alone. Soil moisture, stomata, species and the growing season all matter. Do not infer a unique cause of growth change from a weather change alone.

Working with observations

Record station, elevation, instrument, period, missing values, aggregation and units. Station moves and changes in instrumentation can affect comparisons over time.

Exercise: fictional values

A station recorded 60 mm of precipitation in April and 90 mm in May. What is the total? Do these numbers establish that the nearby forest had no water stress?

Answer: 150 mm. Timing, soil water, temperature and evapotranspiration are unknown, so water stress cannot be determined from these totals alone.

Read a figure: open the US Geological Survey natural water cycle diagram and find precipitation, infiltration and evapotranspiration. The diagram depicts the natural cycle; consider water withdrawals and land-use changes separately.

Continue learning

See Lesson 7 on dendroclimatology, BG09 on climatology and BG03 on mathematics and statistics.

World Meteorological Organization: weather · Japan Meteorological Agency: past observations · US Geological Survey: the water cycle